1.6 Exponentials & Logarithms

Mathematics — OCR A-Level (H240) — Paper 1: Pure Mathematics

Exponential and logarithmic functions describe the mathematics of *proportional change*: a quantity whose instantaneous rate of change is proportional to its current size. This is the natural model for population growth, compound interest, radioactive decay, Newton's law of cooling, charge on a capacitor and countless other physical processes. The chapter builds the pair of inverse functions $y = e^x$ and $y = \ln x$ — together with their general-base relatives $y = a^x$ and $y = \log_a x$ — develops the algebra of logarithm laws, and uses them to solve exponential equations, fit power and exponential laws to experimental data, and model real-world growth and decay. A central idea is that the number $e \approx 2.71828$ is uniquely "natural" because $\frac{d}{dx}e^x = e^x$; every other exponential function carries an awkward factor of $\ln a$.

1. The Exponential Function y = a^x

An *exponential function* has the variable in the exponent rather than the base. This is fundamentally different from a power function like x^2: in $y = 2^x$ the base is fixed and the exponent varies, giving explosive growth; in $y = x^2$ the exponent is fixed and the growth is merely polynomial. By $x = 10$, $x^2 = 100$ but $2^x = 1024$ — already ten times larger, and the gap widens without bound.

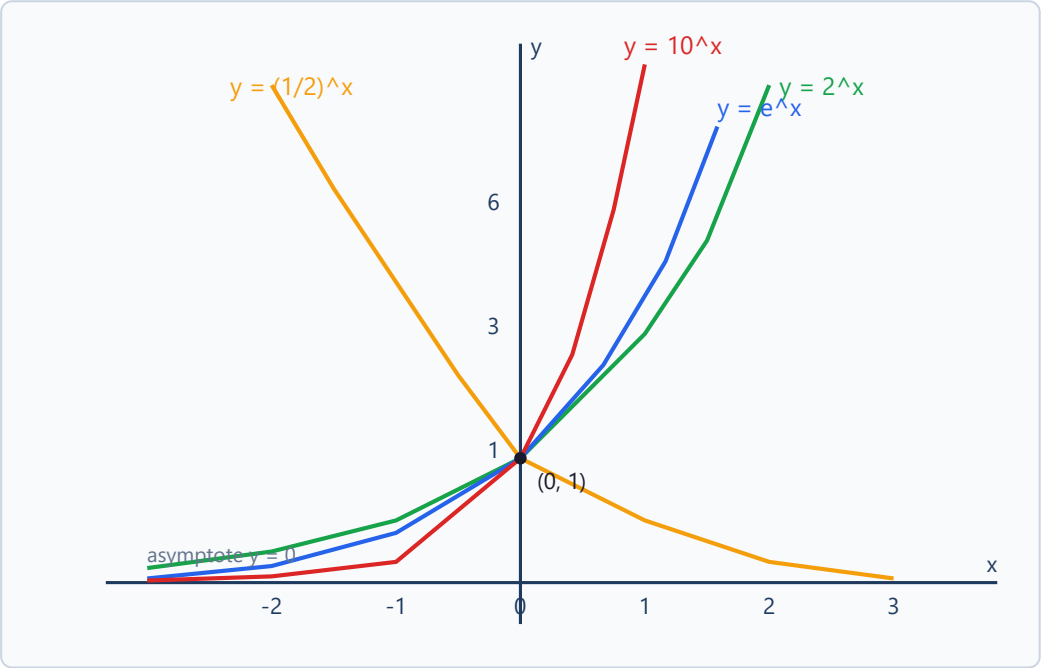
Exponential function. For a constant $a > 0$, $a \neq 1$, the function \[f(x) = a^x, \qquad x \in \mathbb{R} \] is called the exponential function with base a. Its principal properties are:

- **Domain:** all real x.
- **Range:** $(0, \infty)$ — the function is strictly positive for every real x.
- **Intercept:** passes through $(0, 1)$, since $a^0 = 1$ for any $a \neq 0$.
- **Monotonicity:** strictly increasing if $a > 1$; strictly decreasing if $0 < a < 1$.
- **Horizontal asymptote:** $y = 0$ (the x-axis) — as $x \to -\infty$ if $a > 1$, or $x \to +\infty$ if $0 < a < 1$.

- **One-to-one:** so it has an inverse, the logarithm $\log_a$.

Graphs of y = a^x for Various Bases

All exponential curves pass through $(0, 1)$ and approach the x-axis asymptotically. The base controls how steeply they rise or fall. For $a > 1$ they grow faster as a increases; for $a < 1$ they decay faster as a decreases.



Three features are worth noting: (i) all four curves share the point $(0, 1)$; (ii) the curves for $a = 2, e, 10$ are all increasing, with $y = 10^x$ rising fastest and $y = 2^x$ rising slowest; (iii) the curve $y = (1/2)^x = 2^{-x}$ is the reflection of $y = 2^x$ in the y-axis — it decays from $+\infty$ (as $x \to -\infty$) through $(0, 1)$ down towards 0.

Exam tip — exponentials are always positive. Because $a^x > 0$ for every real x and every positive base a, an equation like $e^x = -2$ or $3^x + 1 = 0$ has *no real solution*. Spotting this saves wasted algebra. If after manipulation you end up trying to take $\log$ of a negative number, either there is no solution or there is an error earlier.

2. The Number e

Among all exponential functions $y = a^x$, exactly one has the property that its gradient at every point equals its height — that base is Euler's constant $e \approx 2.71828$. This single fact is the reason e is called the *natural* base: calculus with e^x avoids extra multiplicative factors.

Euler's number e. The irrational constant \[e = 2.71828182845904523536\ldots \] can be defined in several equivalent ways, of which two are standard at A-level: \[e = \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n, \qquad e = \sum_{n=0}^{\infty} \frac{1}{n!} = 1 + 1 + \tfrac{1}{2} + \tfrac{1}{6} + \tfrac{1}{24} + \cdots \] The first (limit) definition arises naturally from continuously compounded interest; the second (series) definition is the Taylor expansion of e^x evaluated at $x = 1$.

Why e is Natural

Differentiating $y = a^x$ from first principles gives \[\frac{dy}{dx} = a^x \cdot \lim_{h \to 0} \frac{a^h - 1}{h}. \] The limit on the right depends only on the base a, not on x. For $a = 2$ it equals $\ln 2 \approx 0.693$; for $a = 10$ it equals $\ln 10 \approx 2.303$; for $a = e$ it equals exactly 1. So e is the unique base for which $\frac{d}{dx}a^x = a^x$ — no awkward factor.

$$\frac{d}{dx}(e^x) = e^x.$$

Compound Interest and the Limit Definition

If £1 is invested at 100% interest compounded n times per year, the year-end value is $(1 + 1/n)^n$. As compounding becomes continuous ($n \to \infty$) this value approaches e. Numerically: with $n = 1$ it is 2; with $n = 12$, 2.613; with $n = 365$, 2.7146; in the limit, e.

n	1	2	12	365	8760	$\to \infty$
$(1 + 1/n)^n$	2.0000	2.2500	2.6130	2.7146	2.71813	e

Exam tip — memorise key values of e^x. $e^0 = 1$; $e^1 \approx 2.718$; $e^2 \approx 7.389$; $e^{-1} \approx 0.3679$; $e^{1/2} \approx 1.6487$. Knowing these speeds up sanity-checks: if your answer involves e^{10} and the question expected a two-digit answer, something is wrong ($e^{10} \approx 22{,}026$).

3. The Natural Logarithm ln(x)

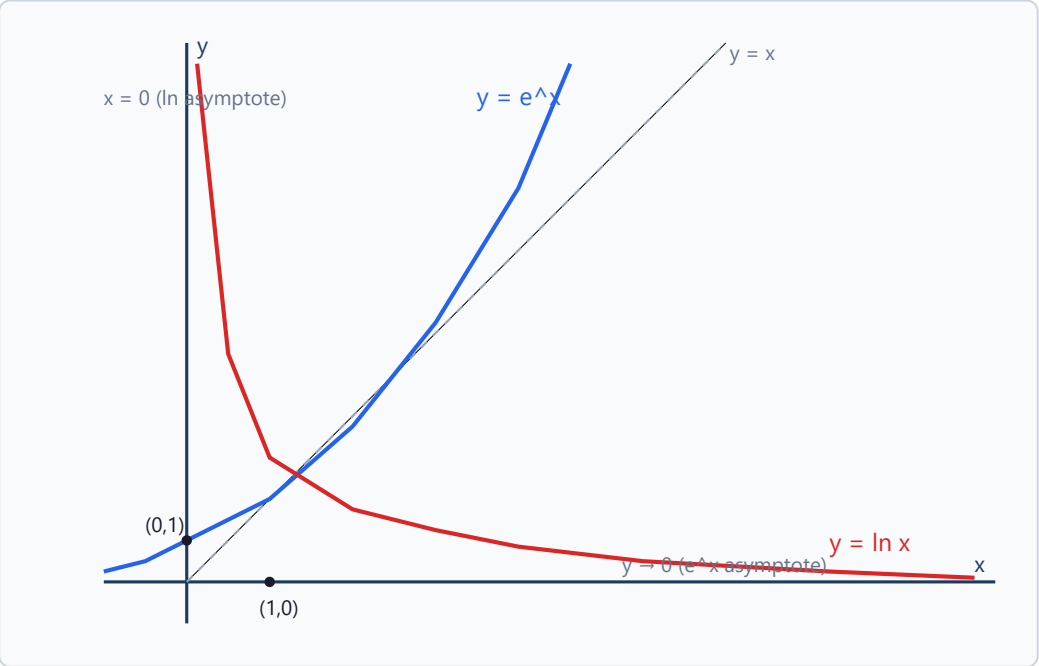
The natural logarithm is the inverse of the exponential function with base e. It answers the question: "to what power must I raise e to get x?"

Natural logarithm. For $x > 0$, \[\ln x = \log_e x \iff e^{\ln x} = x. \] Equivalently, $y = \ln x$ means $e^y = x$. Properties:

- **Domain:** $(0, \infty)$ — cannot take log of zero or a negative number.
- **Range:** all real numbers.
- **Intercept:** passes through $(1, 0)$, since $\ln 1 = 0$.
- **Monotonicity:** strictly increasing.
- **Vertical asymptote:** $x = 0$. As $x \to 0^+$, $\ln x \to -\infty$.
- **Inverse identities:** $\ln(e^x) = x$ for all real x; $e^{\ln x} = x$ for $x > 0$.

Graphs of y = e^x and y = ln x

Because $\ln$ is the inverse of the exponential, their graphs are reflections of each other in the line $y = x$. The exponential rises steeply from the horizontal asymptote; the logarithm rises slowly from the vertical asymptote.



General-Base Logarithm

For any base $a > 0$, $a \neq 1$, the logarithm $\log_a$ is defined by \[y = \log_a x \iff a^y = x. \] Special cases: $\log_{10}$ (written just $\log$ on calculators, the *common logarithm*), $\log_2$ (common in computer science) and $\log_e = \ln$.

Common logarithm. $\log_{10} x$, often written simply $\log x$ on calculators and in engineering, is the logarithm with base 10. It tells you the number's order of magnitude: $\log 1000 = 3$, $\log 10{,}000 = 4$. The decibel and Richter scales are defined in terms of $\log_{10}$.

4. Logarithm Laws

Three laws summarise how logarithms interact with multiplication, division and powers. They turn exponential relationships into linear ones — which is why they were invented in 1614 by John Napier, long before calculus, for the purpose of simplifying astronomical calculation.

Laws of logarithms. For a fixed base $b > 0$, $b \neq 1$, and positive reals x, y: \[\log_b(xy) = \log_b x + \log_b y \qquad \text{(product rule)} \] \[\log_b\!\left(\tfrac{x}{y}\right) = \log_b x - \log_b y \qquad \text{(quotient rule)} \] \[\log_b(x^n) = n \log_b x \qqquad \text{(power rule, any real } n) \] Special values: $\log_b b = 1$; $\log_b 1 = 0$; $\log_b(b^x) = x$; $b^{\log_b x} = x$.

Proof of the Product Rule

Let $\log_b x = p$ and $\log_b y = q$, so by definition $x = b^p$ and $y = b^q$. Then \[xy = b^p \cdot b^q = b^{p + q}, \] using the index law $b^p b^q = b^{p+q}$. Taking $\log_b$ of both sides, \[\log_b(xy) = p + q = \log_b x + \log_b y. \qquad\square \] The quotient rule follows identically using $b^p / b^q = b^{p-q}$; the power rule from $(b^p)^n = b^{np}$.

Change of Base

Change of base. For positive a, b, x with $a, b \neq 1$, \[\log_a x = \frac{\log_b x}{\log_b a} \quad\text{and in particular}\quad \log_a x = \frac{\ln x}{\ln a}. \] *Proof.* Let $y = \log_a x$, so $a^y = x$. Taking $\log_b$ of both sides, $y \log_b a = \log_b x$, which rearranges to the stated formula.

The change-of-base formula means a calculator with only $\ln$ and $\log_{10}$ buttons can compute any logarithm: e.g. $\log_2 5 = \ln 5 / \ln 2 \approx 1.6094/0.6931 \approx 2.322$.

Law	Statement	Typical use
Product	$\log(xy) = \log x + \log y$	Combine two log terms into one; split a log to solve.
Quotient	$\log(x/y) = \log x - \log y$	Simplify $\log(a/b)$ expressions; eliminate a fraction.
Power	$\log(x^n) = n \log x$	Bring an exponent down — essential for solving $a^x = b$.
Zero	$\log_b 1 = 0$	Any log of 1 vanishes — watch for this in simplification.
Unity	$\log_b b = 1$	$\log_{10} 10 = 1$; $\ln e = 1$.
Change of base	$\log_a x = \ln x / \ln a$	Evaluate arbitrary-base logs with a standard calculator.

Exam tip — log laws require positivity. Every log law is stated for *positive* arguments. Writing $\log(x^2) = 2\log x$ is wrong if x could be negative — the left side is defined ($x^2 > 0$) but the right side is not. The safe version is $\log(x^2) = 2\log|x|$. When solving log equations, always check at the end that every argument of every log was positive.

Worked Example 1 — Combining logs

Write $2\ln 3 + \ln 5 - \ln 2$ as a single logarithm.

```
1.  Use power rule on the first term:
2.     2 ln 3 = ln(3²) = ln 9.
3.  Use product rule to combine ln 9 + ln 5:
4.     ln 9 + ln 5 = ln(9·5) = ln 45.
5.  Use quotient rule to subtract ln 2:
6.     ln 45 − ln 2 = ln(45/2) = ln 22.5.
7.  ∴ 2 ln 3 + ln 5 − ln 2 = ln(45/2).

// Numerical check: 2(1.0986) + 1.6094 − 0.6931 = 3.1135 ≈ ln(22.5). ✓
```

5. Solving Exponential Equations

An equation of the form $a^x = b$ or $e^{f(x)} = c$ is solved by taking logs of both sides. Which log? In principle any base works, but in practice use $\ln$ (for $e^{\cdot\cdot\cdot}$) or $\log_{10}$ (for $10^{\cdot\cdot\cdot}$), because every calculator has these. The power rule $\log(a^x) = x \log a$ is what makes this work — it moves the unknown out of the exponent.

Worked Example 2 — Simple exponential

Solve $2^x = 10$.

```
1.  Take ln of both sides (any base works; ln is convenient):
2.     ln(2^x) = ln 10
3.     x · ln 2 = ln 10                (power rule)
4.     x = ln 10 / ln 2.
5.  Numerical value:
6.     x = 2.302585 / 0.693147
7.     x ≈ 3.3219 (4 d.p.)
8.  Check: 2^3.3219 = e^(3.3219 · 0.693) = e^2.303 ≈ 10. ✓

// The exact answer is x = log₂ 10 = ln 10/ln 2 — both are acceptable forms.
```

Worked Example 3 — Different bases on two sides

Solve $3^{2x+1} = 5^x$.

```
1.  Take ln of both sides:
2.     ln(3^(2x+1)) = ln(5^x)
3.     (2x + 1) ln 3 = x ln 5            (power rule)
4.  Expand and gather x-terms:
5.     2x ln 3 + ln 3 = x ln 5
6.     2x ln 3 − x ln 5 = − ln 3
7.     x (2 ln 3 − ln 5) = − ln 3
8.     x = − ln 3 / (2 ln 3 − ln 5) = ln 3 / (ln 5 − 2 ln 3).
9.  Numerically:
10.    x = 1.0986 / (1.6094 − 2.1972) = 1.0986 / (−0.5878) ≈ −1.869.
```

```
// Using the quotient rule: denominator = ln 5 - ln 9 = ln(5/9), so x = ln 3 / ln
```

Worked Example 4 — Exponential plus constant

Solve $e^x + 3 = 5$.

```
1.  Isolate the exponential:
2.     e^x = 5 - 3 = 2.
3.  Take ln:
4.     x = ln 2 ≈ 0.6931.

// Reminder: e^x > 0 always. If we had e^x + 5 = 3, the equation would read
// e^x = -2, which has NO real solution.
```

6. Solving Logarithmic Equations

Log equations are solved by combining terms using the log laws until one side is a single log, then removing the log by exponentiating. Every candidate solution must then be checked: it must make *every* log argument in the original equation positive.

Worked Example 5 — Combined-log equation with domain check

Solve $\log_2 x + \log_2(x-2) = 3$.

```
1.  Domain: need x > 0 AND x - 2 > 0, so x > 2.
2.  Combine using product rule:
3.     log₂(x(x - 2)) = 3
4.     x(x - 2) = 2³ = 8.
5.  Expand:
6.     x² - 2x - 8 = 0
7.     (x - 4)(x + 2) = 0
8.     x = 4   or   x = -2.
```

```
9.  Check domain: x > 2.
10.    x = 4 ✓     x = -2 ✗ (rejected — would give log of negative).
11. ∴ x = 4.

// Forgetting to reject x = -2 is a classic lost mark.
```

Worked Example 6 — Natural-log equation

Solve $\ln x = 2$.

```
1.  By definition of ln, e^(ln x) = x, so exponentiate both sides:
2.     e^(ln x) = e²
3.     x = e² ≈ 7.389.
4.  Domain: x > 0 ✓.
```

7. Modelling with Exponentials

Any quantity whose rate of change is proportional to its current size obeys an exponential law. If $\frac{dP}{dt} = kP$ then $P = P_0 e^{kt}$; positive k gives growth, negative k gives decay. This is the backbone of population models, radioactive decay, compound interest, discharge of a capacitor and heat loss (Newton's law of cooling).

Exponential growth model. \[P(t) = P_0 \, e^{kt}, \qquad k > 0. \] $P_0 = P(0)$ is the initial value; k is the *continuous growth rate*. The *doubling time* T_2 satisfies $P(T_2) = 2P_0$, giving \[T_2 = \frac{\ln 2}{k}. \]

Exponential decay model. \[N(t) = N_0 \, e^{-\lambda t}, \qquad \lambda > 0. \] λ is the *decay constant*. The *half-life* $T_{1/2}$ is the time for the quantity to fall to half its value. From $N_0 e^{-\lambda T_{1/2}} = \frac{1}{2} N_0$: \[T_{1/2} = \frac{\ln 2}{\lambda}. \]

Quantity	Formula	When the variable appears

Doubling time	$T_2 = (\ln 2)/k$	Population, investment growing at rate k.
Half-life	$T_{1/2} = (\ln 2)/\lambda$	Radioactive decay, drug elimination.
Time-to-multiple m	$t = (\ln m)/k$	Population reaches m times its initial size.
Fraction remaining at t	$N/N_0 = e^{-\lambda t} = (1/2)^{t/T_{1/2}}$	Radiocarbon dating, drug levels.

Newton's Law of Cooling

An object at temperature T placed in surroundings at temperature T_s cools at a rate proportional to the excess temperature: \[\frac{dT}{dt} = -k(T - T_s). \] Setting $u = T - T_s$ gives $du/dt = -ku$, so $u = u_0 e^{-kt}$ and \[T(t) = T_s + (T_0 - T_s)\, e^{-kt}. \] At $t = 0$ this gives $T = T_0$; as $t \to \infty$, $T \to T_s$. The graph is a decaying exponential asymptoting to ambient temperature.

Exam tip — read off the model carefully. In $P = P_0 e^{kt}$: P_0 is the initial value (plug in $t = 0$); k is the continuous rate, not the percentage rate. If population grows by 5% per year, then after 1 year $P = 1.05P_0$, so $e^k = 1.05$, giving $k = \ln 1.05 \approx 0.04879$ — *not* 0.05. Always convert between percentage-per-period and continuous rate k with care.

Worked Example 7 — Population doubling

A town of 10,000 people grows at 5% per year. After how many whole years will the population first exceed 20,000?

```
Annual multiplier = 1.05, so P(t) = 10000 · (1.05)^t.
Set P(t) = 20000:
   10000 · (1.05)^t = 20000
   (1.05)^t = 2.
Take ln of both sides:
   t · ln 1.05 = ln 2
   t = ln 2 / ln 1.05 = 0.6931 / 0.04879 ≈ 14.207.
t must be a whole year, so first whole year exceeding is t = 15.
Check:   P(14) = 10000 · 1.05^14 ≈ 19799    (under 20000)
         P(15) = 10000 · 1.05^15 ≈ 20789    (over 20000) ✓
```

```
// Continuous-rate version: k = ln 1.05 ≈ 0.04879; T₂ = ln 2 / k ≈ 14.21 yr.
```

Worked Example 8 — Radiocarbon dating

Carbon-14 has half-life $T_{1/2} = 5730$ years. What fraction of the original carbon-14 remains after 10,000 years?

```
1.  Decay constant from half-life:
2.     λ = ln 2 / T½ = 0.6931 / 5730 ≈ 1.2097 × 10⁻⁴  yr⁻¹.
3.  Fraction remaining after t = 10000 years:
4.     N/N₀ = e^(-λt) = e^(-1.2097e-4 · 10000) = e^(-1.2097).
5.     N/N₀ ≈ 0.2982.
6.  Equivalent form:
7.     N/N₀ = (1/2)^(t/T½) = (1/2)^(10000/5730) = (1/2)^1.7452 ≈ 0.2982. ✓
8.  ∴ about 29.8% remains.

// Radiocarbon dating is reliable up to ~50,000 yr – about 8 half-lives,
// after which the remaining fraction (~0.4%) becomes too small to measure.
```

Worked Example 9 — Newton's law of cooling

A cup of coffee follows $T = 20 + 60\,e^{-0.1t}$, with t in minutes and T in °C. When is $T = 40$°C?

```
1.  Set up the equation:
2.     20 + 60 e^(-0.1 t) = 40
3.     60 e^(-0.1 t) = 20
4.     e^(-0.1 t) = 1/3.
5.  Take ln:
6.     -0.1 t = ln(1/3) = -ln 3
7.     t = (ln 3)/0.1 = 10 ln 3 ≈ 10.986 min.
8.  Initial check: T(0) = 20 + 60 = 80°C ✓; ambient T(s) = 20°C ✓.
```

```
// As t → ∞, T → 20°C (the room temperature). The graph is a decaying
// exponential shifted up by 20.
```

8. Logarithmic Scales — Linearising Data

A key experimental technique is to use logs to convert a non-linear relationship into a linear one. Plotting logged quantities and looking for a straight line lets you read off unknown constants from the intercept and gradient.

Exponential Law y = a·b^x

Take logs of both sides (any base — pick $\log_{10}$ or $\ln$): \[\log y = \log a + x \log b. \] This is linear in x: plotting $\log y$ (vertical) against x (horizontal) gives a straight line with intercept $\log a$ and gradient $\log b$. So an exponential relationship in raw variables becomes a straight line after taking the log of only one axis (semi-log plot).

Power Law y = a·x^n

Take logs: $\log y = \log a + n \log x$. Now plot $\log y$ against $\log x$ — a log-log plot. A straight line with gradient n and intercept $\log a$ confirms a power law and reveals both constants.

Model	Linearised form	Plot	Gradient	Intercept
$y = ab^x$	$\log y = \log a + x \log b$	$\log y$ vs x	$\log b$	$\log a$
$y = ax^n$	$\log y = \log a + n \log x$	$\log y$ vs $\log x$	n	$\log a$
$y = ae^{kx}$	$\ln y = \ln a + kx$	$\ln y$ vs x	k	$\ln a$

Exam tip — plot and check. In OCR modelling questions, if the examiner tells you data "fits a law $y = ab^x$", expect to plot $\log y$ against x and show — using a table of values — that the points are (approximately) collinear. Then fit the line, read off gradient and intercept, and recover a, b by exponentiating. The straight line is the *evidence* that the model is correct; without it, you've only assumed the law.

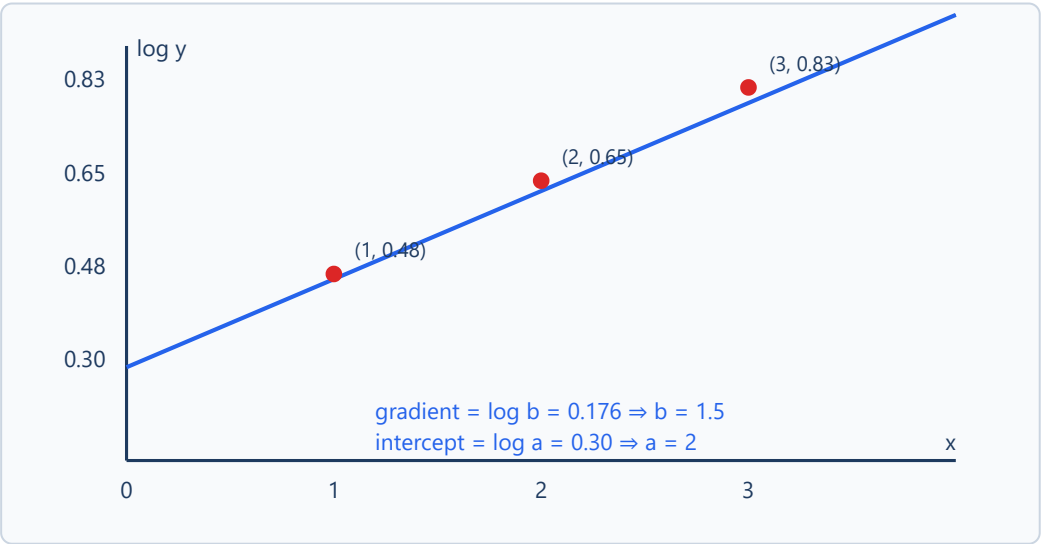
Worked Example 10 — Fitting an exponential to data

Experimental data: $(x, y) = (1, 3.0), (2, 4.5), (3, 6.75)$. Find a and b so that $y = a \cdot b^x$.

```
Linearise: log y = log a + x log b.
Compute log₁₀ of each y:
   x = 1:  log 3.0   = 0.4771
   x = 2:  log 4.5   = 0.6532
   x = 3:  log 6.75  = 0.8293
Check constant first differences (confirming linearity):
   0.6532 − 0.4771 = 0.1761
   0.8293 − 0.6532 = 0.1761    ✓ equal ⇒ exactly linear in x.
Gradient = log b = 0.1761   ⇒   b = 10^0.1761 = 1.5.
Intercept at x = 1:  log a + 0.1761 = 0.4771  ⇒  log a = 0.3010  ⇒  a = 2.
∴ y = 2 · 1.5^x.
Verify:  x = 1:  2 · 1.5 = 3.0 ✓;  x = 2:  2 · 2.25 = 4.5 ✓;  x = 3:  2 · 3.3

// Each y is obtained from the previous by multiplying by 1.5 — a geometric
// progression with common ratio b. Spotting that shortcuts the whole calculation
```

Linearised Plot (Schematic)



9. Differentiation and Integration of Exponentials and Logarithms

The calculus rules for these functions are short, but ubiquitous. Only e^x and $\ln x$ give clean formulas; general-base versions carry an extra factor of $\ln a$.

$$\frac{d}{dx}(e^x) = e^x, \qquad \frac{d}{dx}(e^{kx}) = k\,e^{kx}, \qquad \frac{d}{dx}(a^x) = a^x \ln a.$$

$$\frac{d}{dx}(\ln x) = \frac{1}{x} \quad (x > 0), \qquad \frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}.$$

$$\int e^{kx}\,dx = \frac{1}{k}e^{kx} + C, \qquad \int \frac{1}{x}\,dx = \ln|x| + C.$$

The absolute-value sign in $\ln|x|$ is important: $1/x$ is defined for $x < 0$ as well, and integrating gives $\ln(-x) + C$ there; $\ln|x|$ covers both cases in one formula.

Function	Derivative	Antiderivative
e^x	e^x	$e^x + C$
e^{kx}	ke^{kx}	$(1/k)e^{kx} + C$
a^x	$a^x \ln a$	$a^x / \ln a + C$
$\ln x$	$1/x$	$x \ln x - x + C$ (integration by parts)
$1/x$	$-1/x^2$	$\ln\|x\| + C$
$\log_a x$	$1/(x \ln a)$	$(x \log_a x - x/\ln a) + C$

(The antiderivative of $\ln x$, while listed, is a topic of chapter 1.8 — integration by parts — rather than strict 1.6 content.)

Exam tip — when to use logs to differentiate. For $y = a^x$ or $y = x^x$ or anything with a variable in the exponent, take $\ln$ of both sides first, then differentiate implicitly. For $y = a^x$: $\ln y = x \ln a \Rightarrow$

$y'/y = \ln a \Rightarrow y' = a^x \ln a$. This "logarithmic differentiation" trick converts an exponent-position variable into a multiplier and then a derivative.

10. Quick-Reference Values

x	e^x	$\ln x$	$\log_{10} x$
0.1	1.1052	-2.3026	-1
0.5	1.6487	-0.6931	-0.3010
1	2.7183 ($= e$)	0	0
2	7.3891	0.6931	0.3010
e (≈ 2.718)	15.154	1	0.4343
3	20.086	1.0986	0.4771
5	148.41	1.6094	0.6990
10	22026	2.3026	1
100	2.69×10^{43}	4.6052	2

Two handy identities used throughout: $\ln 2 \approx 0.693$, $\ln 10 \approx 2.303$. And because $\log_{10} x = \ln x / \ln 10$, every $\ln$-to-$\log_{10}$ conversion is a division by 2.303.

Exam tip — the range of a logarithm is all of $\mathbb{R}$. Unlike sine or cosine, the output of $\ln$ or $\log_a$ is not bounded. If an equation reads $\ln x = -15$, the solution $x = e^{-15} \approx 3.06 \times 10^{-7}$ is perfectly valid — *not* "no solution". The only forbidden input is $x \leq 0$; any real value can come *out*.

11. Consolidated Summary and Exam Strategy

Chapter-wide checklist for Exponentials and Logarithms.

- **a^x is always positive.** Rule out solutions where you'd need $a^x \leq 0$.
- **e^x and $\ln x$ are inverses.** $e^{\ln x} = x$ for $x > 0$; $\ln(e^x) = x$ for all x.
- **Three log laws + change of base.** Product, quotient, power — plus $\log_a x = \ln x / \ln a$. These collapse every log expression.
- **Solving $a^x = b$:** take logs, use power rule, divide. Any base works but $\ln$ is default.
- **Solving log equations:** combine to a single log, exponentiate, then *check domain* of every argument.
- **Growth/decay:** $P = P_0 e^{kt}$; doubling time $T_2 = \ln 2/k$; half-life $T_{1/2} = \ln 2/\lambda$.
- **Linearising data:** $y = ab^x$ → plot $\log y$ vs x; $y = ax^n$ → plot $\log y$ vs $\log x$. Gradient gives the power, intercept gives the constant.
- **Calculus:** $(e^{kx})' = ke^{kx}$; $(\ln x)' = 1/x$; $\int e^{kx} dx = e^{kx}/k + C$; $\int dx/x = \ln|x| + C$.
- **State units and check realism** in modelling answers. A population of -200, or a half-life of -3 years, is a sign of arithmetic error.

Typical OCR examiner complaints (from mark reports).

- Writing $\log(x + y) = \log x + \log y$ — wrong: it's the product, not the sum, that splits.
- Failing to reject a log-equation root that makes a log argument negative (e.g. $x = -2$ in Worked Example 5).
- Confusing the percentage growth rate r (e.g. 5%) with the continuous rate k: $k = \ln(1 + r)$, not r.
- Misapplying the power rule as $\log(x^n) = (\log x)^n$ — the exponent multiplies, it does not raise.
- Giving $\ln(-1)$ or $\ln 0$ as a finite number — both are undefined in the reals.
- Dropping the $\ln a$ factor when differentiating a^x for $a \neq e$.
- Integrating $1/x$ as $\ln x + C$ instead of $\ln|x| + C$ and losing marks when the region includes $x < 0$.
- Plotting y against $\log x$ instead of $\log y$ against x for an exponential model, giving a curved graph and no law.

Formula snapshot (memorise).

- Exponential: $y = a^x$, domain $\mathbb{R}$, range $(0, \infty)$, through $(0, 1)$, asymptote $y = 0$.
- Natural exponential: $y = e^x$, with $(e^x)' = e^x$; Euler's constant $e = \lim_{n\to\infty}(1 + 1/n)^n \approx 2.71828$.
- Natural log: $y = \ln x$, inverse of e^x; domain $(0, \infty)$, range $\mathbb{R}$, through $(1, 0)$, asymptote $x = 0$.
- Log laws: $\log(xy) = \log x + \log y$; $\log(x/y) = \log x - \log y$; $\log(x^n) = n \log x$; $\log_a x = \ln x / \ln a$.
- Growth / decay: $P = P_0 e^{kt}$; doubling time $T_2 = \ln 2/k$; half-life $T_{1/2} = \ln 2/\lambda$.
- Linearising: $y = ab^x \Rightarrow \log y = \log a + x \log b$; $y = ax^n \Rightarrow \log y = \log a + n \log x$.
- Calculus: $(e^{kx})' = ke^{kx}$; $(a^x)' = a^x \ln a$; $(\ln x)' = 1/x$; $\int e^{kx} dx = e^{kx}/k + C$; $\int dx/x = \ln|x| + C$.